Prediction of hERG Inhibition by Chemical Large Language Model

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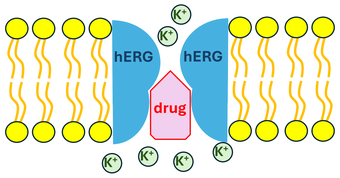
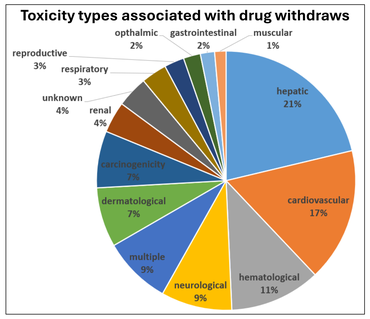
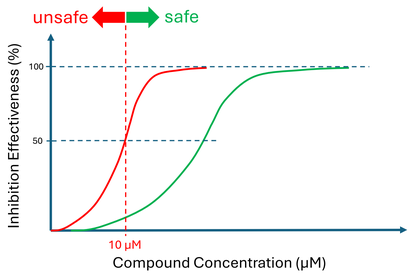
Background

- More than 500 drugs were withdrawn from the market due to **toxicities** [1], where **cardiotoxicity** accounts for 17% of the drug safety issues.
- The **hERG potassium channel inhibition** is the major cause of cardiotoxicity.



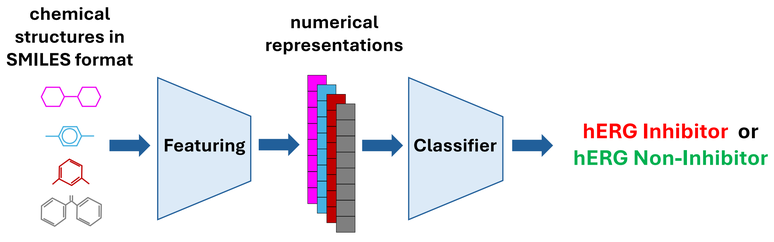


- During drug development, **early identification** of hERG inhibition is critical and can reduce tremendous amount of costs.
- Compounds with **IC50 ≤10 μM** are deemed hERG inhibitor; otherwise hERG non-inhibitor.



- **Problem of assay screening:** Screening tons of compounds is **time consuming**.
- **Solution:** ***In silico* predictive models** can speed up compound screening.

Pipeline of *In Silico* hERG Inhibition Prediction



Experimental Data

- hERG inhibition data of this research is acquired from Therapeutics Data Commons [2, 3].

Training Data

DeepHIT Dataset
- 6,643 hERG Inhibitors
- 5,977 hERG Non-Inhibitors

split by scaffolds

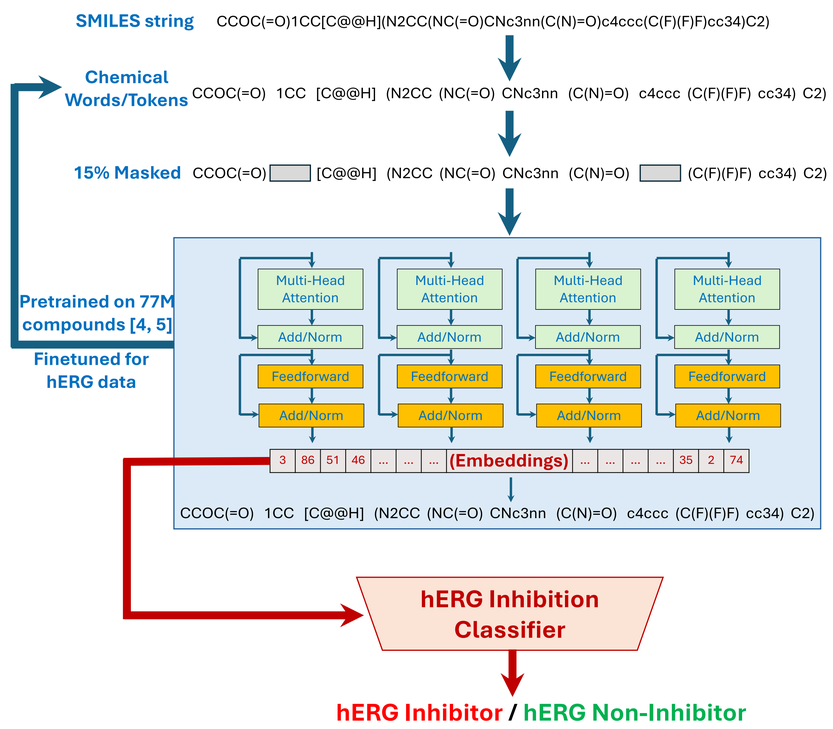
80% → **Training Set**
- 5,338 hERG Inhibitors
- 4,816 hERG Non-Inhibitors

20% → **Validation Set**
- 1,305 hERG Inhibitors
- 1,161 hERG Non-Inhibitors

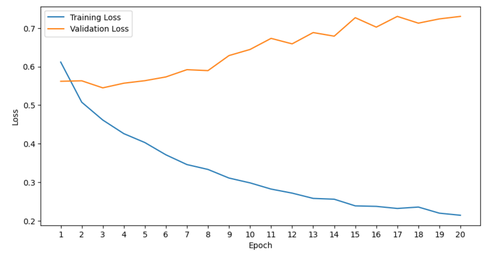
Test Data

External Test-set I	External Test-set II	External Test-set III
• 30 hERG Inhibitors • 14 hERG Non-Inhibitors	• 11 hERG Inhibitors • 30 hERG Non-Inhibitors	• 34 hERG Inhibitors • 706 hERG Non-Inhibitors

Chemical Language Model: ChemBERTa [4, 5]



- The training of the hERG inhibition predictor from the ChemBERTa foundation model can conclude quickly at **epoch #3**.



Test Results

- **Test-set I evaluation:**

Model	Area under ROC Curve	Precision	Recall	F1	Mathews Correlation Coefficient	Specificity
ChemBERTa (this research)	***0.70***	***0.82***	***0.77***	***0.79***	***0.40***	***0.64***
Random Forest on Morgan Descriptor	0.67	0.80	0.70	0.75	0.32	0.64
3-Layer MLP on Morgan Descriptor	0.74	0.85	0.77	0.81	0.46	0.71

- **Test-set II evaluation:**

Model	Area under ROC Curve	Precision	Recall	F1	Mathews Correlation Coefficient	Specificity
ChemBERTa (this research)	***0.74***	***0.43***	***0.91***	***0.59***	***0.42***	***0.57***
Random Forest on Morgan Descriptor	0.69	0.41	0.82	0.55	0.34	0.57
3-Layer MLP on Morgan Descriptor	0.60	0.38	0.55	0.44	0.19	0.67

- **Test-set III evaluation:**

Model	Area under ROC Curve	Precision	Recall	F1	Mathews Correlation Coefficient	Specificity
ChemBERTa (this research)	***0.64***	***0.08***	***0.59***	***0.15***	***0.13***	***0.70***
Random Forest on Morgan Descriptor	0.67	0.08	0.74	0.15	0.14	0.60
3-Layer MLP on Morgan Descriptor	0.69	0.10	0.68	0.17	0.17	0.71

Summary and Future Work

- Chemical large language modeling offers a different framework for molecular feature extraction.
- Features derived from chemical large language modeling can contribute to hERG inhibition prediction.
- Predictor of hERG inhibitor built from the pretrained ChemBERTa model can be trained within a small amount of time.

- Future work:
  - Integrate chemical language features with other featuring approaches to improve hERG inhibition prediction.
  - Apply chemical language modeling to predict other types of toxicities.
  - Develop a generative approach to design drugs without hERG inhibition assisted by chemical language modeling.

References

[1] Siramshetty *et. al.*, WITHDRAWN—a resource for withdrawn and discontinued drugs, *Nucleic Acids Res.*, Vol 44 (2016)

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[4] Chithrananda *et. al.*, ChemBERTa: Large-scale self-supervised pretraining for molecular property prediction, *NeurIPS*, (2020)

[5] Ahmad *et. al.*, ChemBERTa-2: Towards chemical foundation model, *ELLIS Machine Learning for Molecule Discovery Workshop*, (2021)